

JavaScript Fundamentals Worksheet

GreenTech Solutions – Task 2 Preparation

Lesson length: 2 hours

Independent learning: 1 hour

Learning goals (read this first)

By the end of this lesson, you will be able to:

- explain what JavaScript is used for
- connect JavaScript to a web page
- respond to user actions (clicking and typing)
- read values from inputs
- check user input is valid
- update content on the page
- apply JavaScript patterns to a real scenario

You are **not expected to memorise JavaScript**.

You are expected to **understand patterns** and reuse them.

Files you are working with

You have been given a folder containing:

- index.html – the web page structure
- static/style.css – the page styling
- static/app.js – **this is where you work today**
- docs/testing-log.md – record your testing
- readme.md – summary of the project

You will mainly work in **static/app.js**.

Important structure of today's work

Your JavaScript file has **two parts**:

Part 1: JavaScript Practice Area

- Bland and generic
- Used to learn JavaScript safely
- You follow and understand this first

Part 2: GreenTech Application

- Real scenario
- You apply the same ideas
- No new JavaScript ideas are introduced

Rule:

Do not work on Part 2 until you understand Part 1.

PART 1: JavaScript Practice Area (Learning JavaScript)

This section is already written for you.

Your job is to **read it, run it, and understand it**.

Task 1: Number checker (click event)

What this does

- Reads a number from an input box
- Checks it is valid

- Doubles it
- Shows the result on the page

What to do

1. Open index.html in your browser
2. Type a number into the “Enter a number” box
3. Click **Check number**
4. Try:
 - a. leaving it blank
 - b. typing letters
 - c. typing a valid number

Think about:

- What happens when the button is clicked?
- Why does the code use return?
- Where does the output appear?

Task 2: Character counter (input event)

What this does

- Counts characters as the user types
- Updates the number instantly

What to do

1. Type into the “Text box (max 50 characters)”
2. Watch the character count update

Think about:

- Why is this an input event and not a click?
- What does .value.length return?

Stop and check (teacher will pause here)

You should be able to answer:

- What does getElementById do?
- What is an event?
- What is .value?
- Why do we validate input?

If not, ask now.

PART 2: Apply JavaScript to GreenTech

You will now **reuse the same patterns** from Part 1.

You are **not learning new JavaScript** here.

You are **applying what you already saw**.

TASK 2A: GreenTech character counter

Goal

Make the GreenTech forum character counter work.

What you already know

You have already done this in Part 1:

```
demoText.addEventListener("input", function () {  
  demoCharCount.textContent = demoText.value.length;  
});
```

What to do

In **Part 2A** of app.js:

1. Get the forum text box using its id: forumPost
2. Get the counter span using its id: charCount
3. Add an input event listener
4. Update the character count as the user types

Success check

- Type in the GreenTech forum box
- The counter updates correctly

TASK 2B: Clear form button

Goal

Reset the GreenTech form and outputs.

What you already know

- Buttons use `addEventListener("click", ...)`
- Forms can be reset using `.reset()`
- Text is cleared using `.textContent = ""`

What to do

In **Part 2B** of `app.js`:

1. Get:
 - a. the Clear button
 - b. the form
 - c. the message area
 - d. the impact score
 - e. the eco tip
2. Add a click event to the Clear button
3. When clicked:
 - a. reset the form
 - b. clear the message
 - c. reset score and tip text

- d. reset the character counter

Success check

- Click **Clear form**
- Everything resets correctly

TASK 2C: Stop the page refreshing on submit

Goal

Make the form submit without refreshing the page.

What you already know

- Forms refresh by default
- `event.preventDefault()` stops this

What to do

In **Part 2C** of `app.js`:

1. Add a submit event to the GreenTech form
2. Use `event.preventDefault()`
3. Show a simple message to prove it works

Success check

- Click **Calculate impact**
- The page does not refresh
- A message appears

TASK 2D: Read ONE input and show a message

Goal

Read one value from the GreenTech form and show it.

What to do

Choose **one**:

- Name

or

- Travel mode

Steps:

1. Get the input using `getElementById`
2. Read the value using `.value`
3. Display it in `messageArea`

Example outputs

- “Hello Alex”
- “Travel mode selected: bus”

Optional challenge: Basic validation

Goal

Add one validation rule.

Example:

- If the name is blank:
 - show “Please enter your name”
 - stop the code using `return`

Only do this if you are confident.

Testing log (important)

Open docs/testing-log.md and record:

- blank input
- invalid input
- valid input

Testing supports higher marks later.

Independent learning (1 hour)

Choose **one**:

- <https://learnjavascript.online/>
- <https://www.geeksforgeeks.org/javascript/javascript-tutorial/>

Then improve **one thing**:

- better messages
- clearer validation
- improved wording
- small UI improvement

Do not rush. Quality matters more than quantity.

Reflection (complete at the end)

Answer in full sentences:

1. What is JavaScript doing on this page that HTML cannot do alone?
2. Which pattern did you reuse from the practice area?
3. What went wrong at first, and how did you fix it?
4. What would you improve next?

Final reminder

You are learning **how JavaScript works**, not copying answers.

If you can explain your code, you are doing it right.